Exp No: 1	Study of various Network commands used in Linux and Windows

AIM:

To study and practise the various network commands used in the Windows and Linux operating systems.

ALGORITHM / PROCEDURE:

1. Open the Command Prompt (`cmd`) in Windows and the Terminal in Linux.
2. Use `hostname` to see the name of the computer.
3. Use `ipconfig /all` (Windows) and `ip address show` (Linux) to see the IP address, subnet mask, MAC address and DHCP details.
4. Use `ping` to check whether a remote system is reachable and note the packet loss and the time taken.
5. Use `tracert` (Windows) and `traceroute` (Linux) to see the path taken by the packets.
6. Use `nslookup` to find the IP address of a domain name.
7. Use `netstat -r` and `route print` to see the routing table, and `ip route` in Linux.
8. Use `arp -a` to see the IP address to MAC address table.
9. Note down the output of each command.

Commands used:

Purpose	Windows	Linux
Host name	hostname	hostname
IP configuration	ipconfig /all	ip address show
Check connectivity	ping google.com	ping google.com
Trace the route	tracert google.com	traceroute google.com
DNS lookup	nslookup www.google.com	nslookup / dig
ARP table	arp -a	arp -a
Routing table	netstat -r, route print	ip route

DIAGRAM:

Fig 1: Network commands and the layers they work in

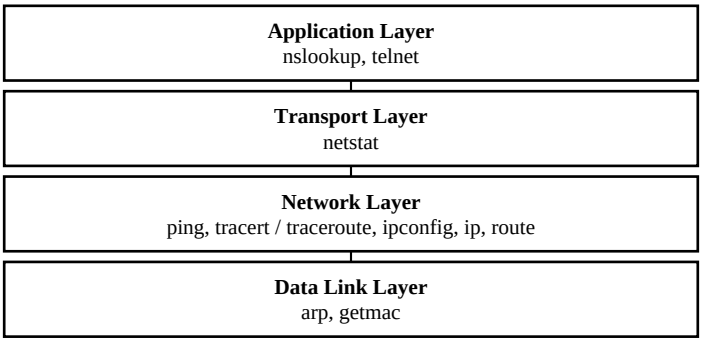
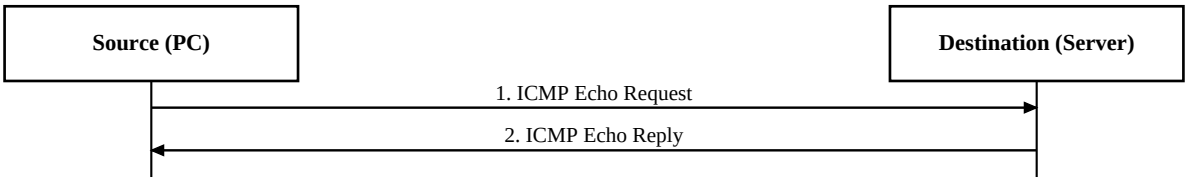


Fig 2: Working of the ping command



OUTPUT:

Windows:

Command Prompt

```
C:\Users\Darshan Abinav RK>hostname
Darshan Abinav RK
```

Command Prompt

```
C:\Users\Darshan Abinav RK>ipconfig /all
Windows IP Configuration
 Host Name . . . . . . . . . . . . : Darshan Abinav RK
 Primary Dns Suffix . . . . . . . :
 Node Type . . . . . . . . . . . . : Mixed
 IP Routing Enabled. . . . . . . . : No
 WINS Proxy Enabled. . . . . . . . : No
Ethernet adapter Ethernet 2:
 Connection-specific DNS Suffix . :
 Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
 Physical Address. . . . . . . . . : 0A-00-27-00-00-0E
 DHCP Enabled. . . . . . . . . . . : No
 Autoconfiguration Enabled . . . . : Yes
 Link-local IPv6 Address . . . . . : fe80::aa4a:775:1de1:1757%14(Preferred)
 IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
 Subnet Mask . . . . . . . . . . . : 255.255.255.0
 Default Gateway . . . . . . . . . :
 DHCPv6 IAID . . . . . . . . . . . : 118095911
 DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F
 NetBIOS over Tcpip. . . . . . . . : Enabled
Wireless LAN adapter Local Area Connection* 1:
 Media State . . . . . . . . . . . : Media disconnected
 Connection-specific DNS Suffix . :
 Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
 Physical Address. . . . . . . . . : F8-FE-5E-3A-70-56
 DHCP Enabled. . . . . . . . . . . : Yes
 Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Local Area Connection* 2:
 Media State . . . . . . . . . . . : Media disconnected
 Connection-specific DNS Suffix . :
 Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
 Physical Address. . . . . . . . . : FA-FE-5E-3A-70-55
 DHCP Enabled. . . . . . . . . . . : No
 Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Wi-Fi:
 Connection-specific DNS Suffix . :
 Description . . . . . . . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
 Physical Address. . . . . . . . . : F8-FE-5E-3A-70-55
 DHCP Enabled. . . . . . . . . . . : Yes
 Autoconfiguration Enabled . . . . : Yes
 Link-local IPv6 Address . . . . . : fe80::b810:29c9:55e9:3a65%17(Preferred)
 IPv4 Address. . . . . . . . . . . : 10.83.96.83(Preferred)
 Subnet Mask . . . . . . . . . . . : 255.255.255.0
 Lease Obtained. . . . . . . . . . : 26 August 2026 14:38:14
 Lease Expires . . . . . . . . . . : 26 August 2026 15:38:11
 Default Gateway . . . . . . . . . : 10.83.96.20
 DHCP Server . . . . . . . . . . . : 10.83.96.20
 DHCPv6 IAID . . . . . . . . . . . : 167312990
```

Command Prompt

```
 DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F
 DNS Servers . . . . . . . . . . . : 10.83.96.20
 NetBIOS over Tcpip. . . . . . . . : Enabled
Ethernet adapter Bluetooth Network Connection:
 Media State . . . . . . . . . . . : Media disconnected
 Connection-specific DNS Suffix . :
 Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
 Physical Address. . . . . . . . . : F8-FE-5E-3A-70-59
 DHCP Enabled. . . . . . . . . . . : Yes
 Autoconfiguration Enabled . . . . : Yes
Ethernet adapter Ethernet:
 Media State . . . . . . . . . . . : Media disconnected
 Connection-specific DNS Suffix . :
 Description . . . . . . . . . . . : Realtek Gaming GbE Family Controller
 Physical Address. . . . . . . . . : 2C-58-B9-DE-5F-5F
 DHCP Enabled. . . . . . . . . . . : Yes
 Autoconfiguration Enabled . . . . : Yes
```

Command Prompt

```
C:\Users\Darshan Abinav RK>arp -a
Interface: 192.168.56.1 --- 0xe
 Internet Address      Physical Address              Type
 192.168.56.255        ff-ff-ff-ff-ff-ff     static
 224.0.0.22            01-00-5e-00-00-16             static
 224.0.0.251           01-00-5e-00-00-fb             static
 224.0.0.252           01-00-5e-00-00-fc         static
 239.255.255.250       01-00-5e-7f-ff-fa             static
 255.255.255.255       ff-ff-ff-ff-ff-ff     static
Interface: 10.83.96.83 --- 0x11
 Internet Address      Physical Address              Type
 10.83.96.20           4e-ec-85-ec-ff-15         dynamic
 10.83.96.255          ff-ff-ff-ff-ff-ff     static
 224.0.0.22            01-00-5e-00-00-16             static
 224.0.0.251           01-00-5e-00-00-fb             static
 224.0.0.252           01-00-5e-00-00-fc         static
 239.255.255.250       01-00-5e-7f-ff-fa             static
 255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Command Prompt

```
C:\Users\Darshan Abinav RK>netstat -r
===========================================================================
Interface List
14...0a 00 27 00 00 0e ......VirtualBox Host-Only Ethernet Adapter
11...f8 fe 5e 3a 70 56 ......Microsoft Wi-Fi Direct Virtual Adapter
 4...fa fe 5e 3a 70 55 ......Microsoft Wi-Fi Direct Virtual Adapter #2
17...f8 fe 5e 3a 70 55 ......Intel(R) Wi-Fi 6E AX211 160MHz
19...f8 fe 5e 3a 70 59 ......Bluetooth Device (Personal Area Network)
 2...2c 58 b9 de 5f 5f ......Realtek Gaming GbE Family Controller
 1...........................Software Loopback Interface 1
===========================================================================
IPv4 Route Table
===========================================================================
Active Routes:
Network Destination           Netmask         Gateway         Interface Metric
      0.0.0.0       0.0.0.0      10.83.96.20       10.83.96.83     35
    10.83.96.0 255.255.255.0             On-link      10.83.96.83 291
   10.83.96.83 255.255.255.255             On-link        10.83.96.83 291
  10.83.96.255 255.255.255.255             On-link         10.83.96.83 291
    127.0.0.0        255.0.0.0        On-link        127.0.0.1 331
    127.0.0.1 255.255.255.255             On-link        127.0.0.1 331
 127.255.255.255 255.255.255.255               On-link        127.0.0.1 331
  192.168.56.0 255.255.255.0             On-link        192.168.56.1 281
  192.168.56.1 255.255.255.255               On-link       192.168.56.1 281
 192.168.56.255 255.255.255.255               On-link      192.168.56.1 281
    224.0.0.0        240.0.0.0        On-link        127.0.0.1 331
    224.0.0.0        240.0.0.0        On-link     192.168.56.1 281
    224.0.0.0        240.0.0.0        On-link       10.83.96.83 291
 255.255.255.255 255.255.255.255               On-link        127.0.0.1 331
 255.255.255.255 255.255.255.255               On-link       192.168.56.1 281
 255.255.255.255 255.255.255.255         On-link   10.83.96.83 291
===========================================================================
Persistent Routes:
 None
IPv6 Route Table
===========================================================================
Active Routes:
If Metric Network Destination       Gateway
 1 331 ::1/128            On-link
14 281 fe80::/64              On-link
17 291 fe80::/64              On-link
14 281 fe80::aa4a:775:1de1:1757/128
                    On-link
17 291 fe80::b810:29c9:55e9:3a65/128
                    On-link
 1 331 ff00::/8           On-link
14 281 ff00::/8             On-link
17 291 ff00::/8             On-link
===========================================================================
Persistent Routes:
 None
```

Command Prompt

```
C:\Users\Darshan Abinav RK>nslookup www.google.com
Server: UnKnown
Address: 10.83.96.20
Non-authoritative answer:
Name: www.google.com
Addresses: 2001:4860:482d:7700::
     2001:4860:4826:7700::
     2001:4860:4828:7700::
     2001:4860:482b:7700::
     2001:4860:4827:7700::
     2001:4860:4829:7700::
     2001:4860:482a:7700::
     2001:4860:482c:7700::
     142.251.152.119
     142.251.156.119
     142.251.154.119
     142.251.151.119
     142.251.153.119
     142.251.157.119
     142.251.150.119
     142.251.155.119
```

Command Prompt

```
C:\Users\Darshan Abinav RK>ping google.com
Pinging google.com [142.251.43.46] with 32 bytes of data:
Reply from 142.251.43.46: bytes=32 time=14ms TTL=114
Reply from 142.251.43.46: bytes=32 time=233ms TTL=114
Reply from 142.251.43.46: bytes=32 time=59ms TTL=114
Reply from 142.251.43.46: bytes=32 time=328ms TTL=114
Ping statistics for 142.251.43.46:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 14ms, Maximum = 328ms, Average = 158ms
```

Command Prompt

```
C:\Users\Darshan Abinav RK>route print
===========================================================================
Interface List
14...0a 00 27 00 00 0e ......VirtualBox Host-Only Ethernet Adapter
11...f8 fe 5e 3a 70 56 ......Microsoft Wi-Fi Direct Virtual Adapter
 4...fa fe 5e 3a 70 55 ......Microsoft Wi-Fi Direct Virtual Adapter #2
17...f8 fe 5e 3a 70 55 ......Intel(R) Wi-Fi 6E AX211 160MHz
19...f8 fe 5e 3a 70 59 ......Bluetooth Device (Personal Area Network)
2...2c 58 b9 de 5f 5f ......Realtek Gaming GbE Family Controller
 1...........................Software Loopback Interface 1
===========================================================================
IPv4 Route Table
===========================================================================
Active Routes:
Network Destination           Netmask         Gateway         Interface Metric
0.0.0.0       0.0.0.0      10.83.96.20      10.83.96.83      35
    10.83.96.0 255.255.255.0              On-link      10.83.96.83 291
   10.83.96.83 255.255.255.255              On-link       10.83.96.83 291
   10.83.96.255 255.255.255.255              On-link         10.83.96.83 291
127.0.0.0        255.0.0.0        On-link         127.0.0.1 331
    127.0.0.1 255.255.255.255              On-link        127.0.0.1 331
 127.255.255.255 255.255.255.255                 On-link        127.0.0.1 331
192.168.56.0 255.255.255.0              On-link      192.168.56.1 281
  192.168.56.1 255.255.255.255              On-link       192.168.56.1 281
 192.168.56.255 255.255.255.255                 On-link    192.168.56.1 281
224.0.0.0        240.0.0.0       On-link       127.0.0.1 331
    224.0.0.0        240.0.0.0       On-link     192.168.56.1 281
224.0.0.0        240.0.0.0       On-link      10.83.96.83 291
 255.255.255.255 255.255.255.255              On-link      127.0.0.1 331
 255.255.255.255 255.255.255.255              On-link    192.168.56.1 281
255.255.255.255 255.255.255.255              On-link     10.83.96.83 291
===========================================================================
Persistent Routes:
 None
IPv6 Route Table
===========================================================================
Active Routes:
If Metric Network Destination         Gateway
1 331 ::1/128              On-link
14 281 fe80::/64                On-link
17 291 fe80::/64                On-link
14 281 fe80::aa4a:775:1de1:1757/128
                     On-link
17 291 fe80::b810:29c9:55e9:3a65/128
On-link
 1 331 ff00::/8             On-link
14 281 ff00::/8             On-link
17 291 ff00::/8             On-link
===========================================================================
Persistent Routes:
 None
```

Command Prompt

```
C:\Users\Darshan Abinav RK>tracert google.com

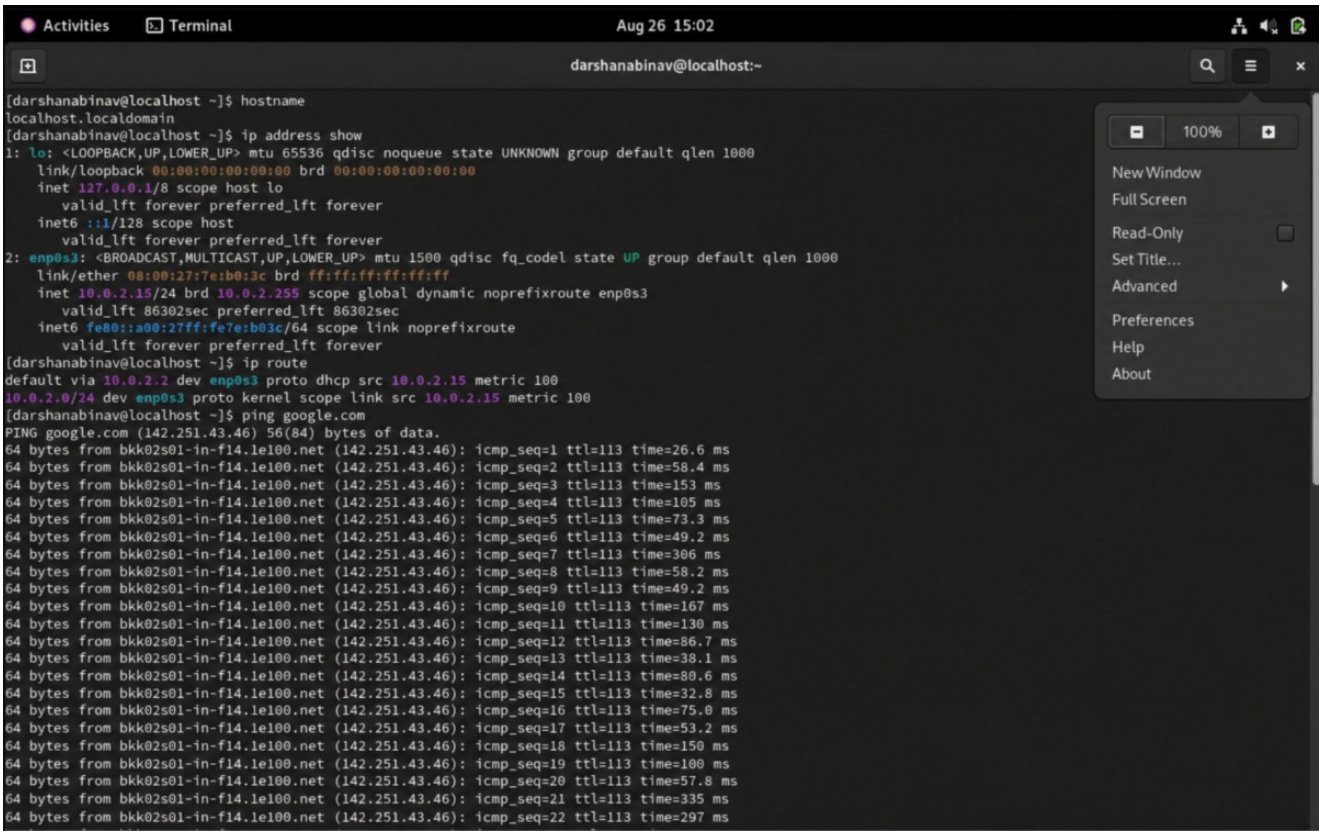
Tracing route to google.com [142.250.195.78]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  192.168.56.1
  2     2 ms     2 ms     2 ms  10.83.96.1
  3     7 ms     6 ms     7 ms  10.83.96.83
  4    13 ms    12 ms    12 ms  106.51.28.57
  5    15 ms    15 ms    14 ms  106.51.28.58
  6    22 ms    21 ms    22 ms  203.50.11.241
  7    24 ms    23 ms    24 ms  72.14.205.193
  8    28 ms    27 ms    27 ms  209.85.243.33
  9    34 ms    33 ms    33 ms  142.250.217.29
 10    35 ms    34 ms    35 ms  142.250.195.78

Trace complete.

C:\Users\Darshan Abinav RK>
```

Linux:



RESULT:

Thus the various network commands used in Windows and Linux were studied and executed successfully.